Hydraulic and Kinematic Systems in Historical Jordan: Three Case Studies from the Nabataean to the Ottoman Period

A Historical–Engineering Review

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Abstract

Mechanical engineering is usually presented as a discipline that took shape in the eighteenth and nineteenth centuries. Yet communities in what is now Jordan addressed problems of the same kind much earlier: moving water through arid terrain, converting rotary motion into a reciprocating cutting stroke, and converting the energy of flowing water into milling power. This review examines three cases drawn from nearly two millennia of the region's history: the water-supply system of Nabataean Petra (1st century BCE–3rd century CE), a sixth-century water-powered stone sawmill at Jerash, and the Mamluk- and Ottoman-period watermills of the Ajloun–Kufranjah valley (Wadi al-Tawaheen). Each case is read against one modern principle: free-surface critical flow and conduit hydraulics, the slider-crank mechanism, and rotational power transmission. The cases are treated as separate examples rather than as evidence of a single continuous tradition, and the strength of the available evidence is graded for each. Petra is the best documented but relies heavily on one research programme; the Jerash mill is supported by excavation but not by published dimensions or efficiency data; and the Ajloun mills are documented mainly in non-peer-reviewed heritage literature. The review identifies concrete research gaps and argues that these sites are useful teaching cases for fluid mechanics, kinematics, and energy conversion.

Keywords: history of engineering, Nabataean hydraulics, Petra, slider-crank mechanism, Jerash, watermills, Jordan, fluid mechanics, kinematics

1. Introduction

Engineering students are often taught that the discipline began with the Scientific and Industrial Revolutions, when thermodynamics, strength of materials, and kinematics were formalized. That account is incomplete. Long before these theories were written down, builders produced water-distribution systems whose behaviour is consistent with hydraulic principles, mechanisms that convert rotation into reciprocating motion, and mills that transmit power from a waterwheel to a millstone. This paper examines whether such artefacts can be meaningfully analysed with modern engineering concepts, and what that analysis can and cannot show.

Jordan offers a useful setting because Nabataean, Roman, Byzantine, and Islamic-era water systems and mills are preserved within a small geographic area. Three sites were selected because each has published documentation relevant to a distinct mechanical principle: Petra, Jerash, and the Ajloun–Kufranjah valley. The selection is purposive, not exhaustive, and is described in Section 3.

The aims of this review are (i) to summarize the literature on these three sites, (ii) to identify the mechanical principle each illustrates, (iii) to grade the strength of the evidence, and (iv) to outline research gaps and educational uses. The review does not claim that the builders understood or applied modern theory. At most, their designs are consistent with it, and the question of intent is addressed explicitly in the discussion.

2. Background and Literature Review

Petra. Ortloff has applied computational fluid dynamics (CFD) to components of Petra's water-supply system, including the Wadi Mataha pipeline system, to assess whether the gradients are consistent with rational hydraulic design rather than only with terrain constraints [1], [2]. Both cited papers are by the same author, so they are not independent confirmation of one another. Independent validation of this CFD work is not currently available in the literature reviewed here.

Jerash. Seigne reported a sixth-century water-powered stone sawmill in the Temple of Artemis precinct and discussed its place in the city's water management [3]. Comparable installations are known elsewhere in the Roman and Byzantine world; a third-century relief from Hierapolis is commonly cited as showing a crank-and-connecting-rod saw drive [4]. The Jerash mill should therefore be read as a regional example of a wider technology, not as a local invention.

Levantine watermills. Schriwer's doctoral dissertation is the principal scholarly synthesis of mills, irrigation, and

agriculture in Bilād al-Shām, 1100–1850 CE [5]. The Ajloun–Kufranjah valley (Wadi al-Tawaheen) has been documented mainly in heritage-management material [6], which is not peer-reviewed and is rarely written from a mechanical-engineering perspective.

3. Approach and Evidence Grading

This is a narrative, purposive review, not a systematic one, and no claim of exhaustive coverage is made. It draws on the principal published sources available for each site. Sources were included if they (i) concern one of the three sites, or a directly comparable installation used for context; (ii) provide technical, archaeological, or documentary data; and (iii) are available in English. Promotional material and sources without identifiable authorship or provenance were not used.

Each case is graded on a three-level scale for the strength of its evidence:

Grade	Meaning
A	Excavated or surveyed physical evidence plus multiple independent peer-reviewed technical analyses
B	Peer-reviewed publication(s), but from a single research programme, or physical evidence without technical parameters
C	Mainly non-peer-reviewed or secondary documentation, or a mechanical interpretation that is inferred

4. Case Study 1: Water Supply at Petra (1st Century BCE–3rd Century CE)

4.1 System description

Petra lies in a semi-arid region, and its water supply combined springs, cisterns, dams, rock-cut channels, and terracotta pipelines. Water from sources such as Ain Musa was conveyed toward the city through conduits that included the Siq and secondary systems such as Wadi Mataha [1], [2]. Because the system combined open channels and closed pipelines, both free-surface and full-pipe flow descriptions are relevant, and they must not be conflated.

In this paper, *channel* means a free-surface conduit, *pipeline* means a closed conduit (terracotta pipes), and *conduit* is used for either.

4.2 Mechanical principles

The appropriate analytical tool depends on the flow condition.

Free-surface (open-channel) flow. The Froude number is

$$\mathrm{Fr} = v / \sqrt{(g \cdot D)} \tag{1}$$

where v is mean velocity, g is gravitational acceleration, and D = A/T is the hydraulic depth (flow area over free-surface width). Flow is subcritical for Fr < 1, critical for Fr = 1, and supercritical for Fr > 1. The specific energy is

$$E = y + v^2 / (2g) \tag{2}$$

where y is the flow depth (small bed slope, kinetic-energy coefficient taken as 1). At critical flow, the specific energy is a minimum for a given discharge Q or, equivalently, the discharge is a maximum for a given specific energy. Critical flow is the boundary between the subcritical and supercritical regimes.

Critical slope. For a given discharge and roughness, the *critical slope* S_c is the bed slope at which uniform flow occurs at critical depth y_c (Fr = 1). Using Manning's equation,

$$S_c = n^2 Q^2 / (A^2 R^{4/3}), \text{ evaluated at } y = y_c \tag{3}$$

where n is Manning's roughness coefficient, A the flow area, and R the hydraulic radius. S_c depends on discharge, channel shape, and roughness, so a slope value is meaningful only together with those parameters.

Pressurized (full-pipe) flow. The Froude number is not defined for a full pipe. Head loss is described with the Darcy–Weisbach equation:

$$h_f = f \cdot (L/d) \cdot v^2 / (2g) \tag{4}$$

where f is the friction factor, L the pipe length, and d the internal diameter. Claims of favourable performance for a terracotta pipeline should be framed in terms of pressure head, head loss, and joint integrity, not Froude number.

Illustrative calculation (hypothetical inputs, not Petra data). Figure 1 applies Eqs. (1)–(3) to a rectangular channel with width b = 0.30 m, Manning coefficient n = 0.015 s/m$^{1/3}$, and discharge Q = 0.040 m³/s. The critical depth is $y_c = (Q^2/(g\,b^2))^{1/3} \approx 0.122$ m, and the corresponding critical slope is $S_c \approx 0.98\%$ (about 0.56°). A bed slope of 0.2% gives subcritical uniform flow (y ≈ 0.22 m, v ≈ 0.60 m/s, Fr ≈ 0.40), whereas a slope of 3% gives supercritical flow (y ≈ 0.082 m, v ≈ 1.63 m/s, Fr ≈ 1.82). The critical slope therefore depends on the discharge and the channel, and a different Q, width, or roughness shifts it.

What the Petra literature reports. Ortloff's CFD work is interpreted as showing that conduit gradients in parts of the system are consistent with hydraulically favourable operation [1], [2]. Whether a particular reported gradient refers to an open channel or a pipeline, and under what assumed discharge and roughness, determines which of the relations above applies; these assumptions should be consulted in the original papers before any single slope value is quoted.

Intent. Agreement between an ancient gradient and a modern optimum does not by itself demonstrate design intent. Plausible explanations include terrain constraints, iterative trial and error, and maintenance feedback over decades (leaks, silting, overflow). The Nabataeans did not have this mathematical description; the relevant claim is only that the observed configuration is consistent with it.

4.3 Critical assessment and evidence grade

Petra is the best documented of the three cases, with peer-reviewed publications in water-engineering and archaeological-science journals [1], [2]. However, the central interpretive claims come from one research programme. CFD reconstructions are sensitive to assumptions that cannot be directly observed today, such as original pipe dimensions, wall roughness, joint leakage, sediment, and inflow conditions. Independent validation, such as experiments on replica pipe sections or in-situ flow measurements, would considerably strengthen the case.

Evidence grade: B (peer-reviewed and detailed, but dominated by a single research programme and not experimentally validated).

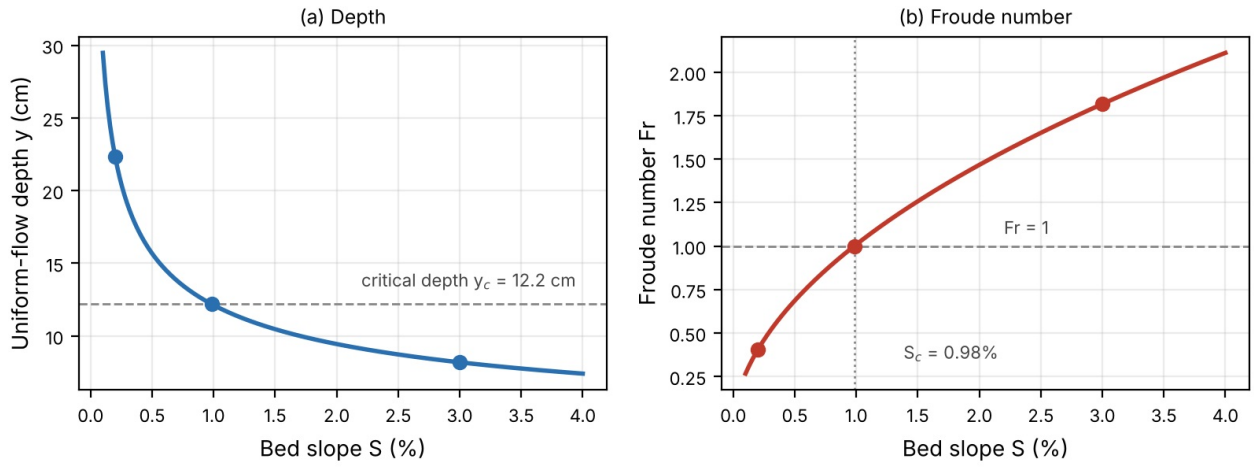


Figure 1. Uniform-flow depth (a) and Froude number (b) as functions of bed slope for a hypothetical rectangular channel (Manning's equation; b = 0.30 m, n = 0.015 s/m$^{1/3}$, Q = 0.040 m³/s). Illustrative calculation by the author; the inputs are chosen for illustration and are not Petra data.

5. Case Study 2: The Water-Powered Sawmill at Jerash (6th Century CE)

5.1 System description

Seigne reported a water-powered sawmill for cutting stone in the Temple of Artemis precinct at Jerash, studied in connection with the city's water-supply infrastructure [3]. Machines of this type use a waterwheel to turn a shaft whose crank, via a connecting rod, drives a reciprocating saw frame. Parallel evidence from Hierapolis [4] indicates that the Jerash mill belongs to a broader Roman–Byzantine tradition. As with comparable installations, the distinction between surviving remains and reconstruction by analogy should be kept in mind when interpreting the mechanism.

5.2 Mechanical principle: the slider-crank mechanism

For an in-line slider-crank (slider line passing through the crank axis, with no offset), the slider position measured from the crank axis is

$$x(\theta) = r\cos\theta + \sqrt{(l^2 - r^2\sin^2\theta)} \tag{5}$$

where r is the crank radius, l the connecting-rod length, and θ the crank angle. The stroke is 2r. Differentiating with $\theta = \omega t$ gives the slider velocity:

$$v(\theta) = dx/dt = -r\,\omega\sin\theta \cdot [1 + (r\cos\theta) / \sqrt{(l^2 - r^2\sin^2\theta)}] \tag{6}$$

For $r/l \ll 1$ this reduces to $v \approx -r\,\omega\,(\sin\theta + (r/2l)\sin 2\theta)$. The motion is therefore nearly sinusoidal, with higher harmonics that grow with r/l. Because the saw frame must reverse at each end of the stroke, the cutting speed varies throughout the stroke and is zero at the dead centres.

Illustrative example (hypothetical values, not site data). Take r = 0.25 m and ω = 1 rad/s. The stroke is 2r = 0.50 m, the peak slider speed is approximately rω = 0.25 m/s for small r/l, and the mean speed over a stroke is $2r\omega/\pi \approx 0.16$ m/s. Doubling ω doubles both. Figure 2 shows the general arrangement of such a saw installation.

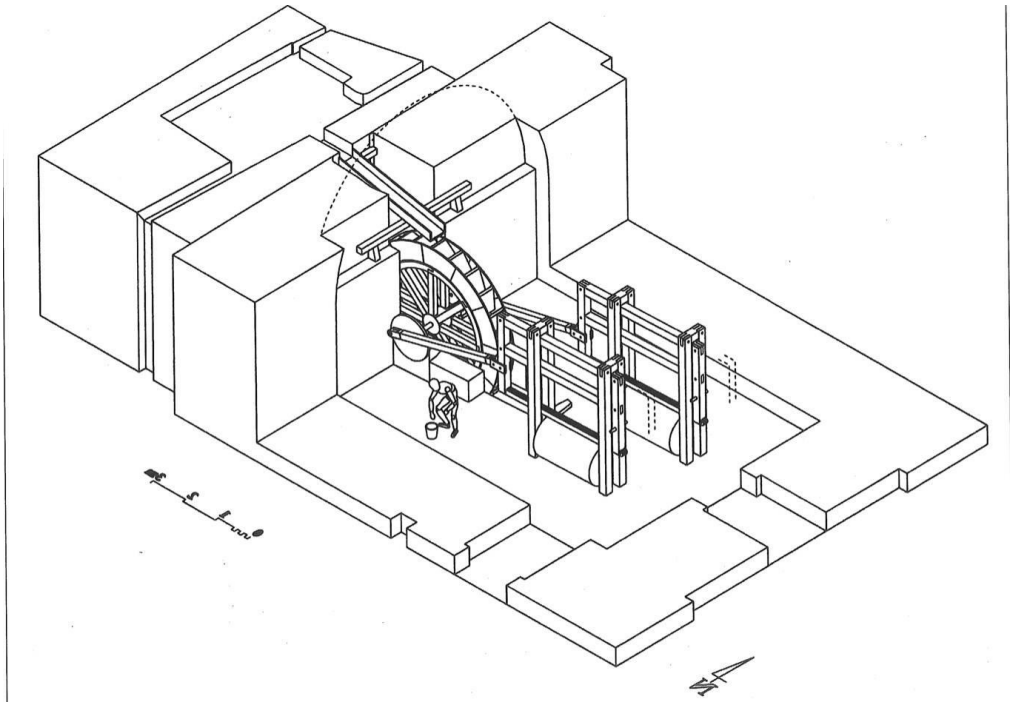


Figure 2. Isometric reconstruction of a Roman/Byzantine water-powered stone-sawing installation, showing the waterwheel, the offset crank, and the frame-mounted saw blades driven in reciprocating motion. General type; not a site-specific reconstruction of the Jerash remains. Source: [SOURCE AND PERMISSION TO BE ADDED].

5.3 Critical assessment and evidence grade

The existence of the mill is documented by excavation [3]. Technical parameters of the Jerash mill (crank radius, rod length, wheel size, head, rotation rate, saw speed, power) are not available in the sources reviewed, and no efficiency value for the mill has been published. Efficiency data for Roman vertical-wheel mills in general are discussed in the literature on Roman water-mills [7], [8], but these figures cannot be transferred to Jerash without justification. Because the Hierapolis evidence is earlier than the Jerash mill [4], Jerash should be described as an example of the slider-crank in use, not as an early or first instance.

Evidence grade: B for the existence of the mill, C for its mechanical detail.

6. Case Study 3: Traditional Watermills of the Ajloun–Kufranjah Valley

6.1 System description

The Ajloun–Kufranjah tributaries (Wadi al-Tawaheen, “Valley of the Mills”) contain a number of watermills, many of them attributed to the Mamluk and Ottoman periods [5], [6]. They were built of local stone, and water was delivered

to the wheel by a channel or tower; the wheel then drove a millstone either directly or through gearing.

Wheel type. Levantine mills are described in the literature as commonly using horizontal wheels on a vertical shaft, fed from a tall tower or penstock, so that the wheel drives the millstone directly without gears [5]. The wheel type at the individual Ajloun mills is not established in the sources reviewed and is treated here as an open question. The relations below apply to water-driven wheels in general; Figure 3 shows the overshot wheel, the configuration for which they are most commonly introduced.

6.2 Mechanical principle: energy conversion and power transmission

For any water-driven wheel, the hydraulic power available from a discharge Q falling through an effective head H is

$$P_{hyd} = \rho\, g\, Q\, H \tag{7}$$

where ρ is water density and g gravitational acceleration. The shaft power is

$$P_{shaft} = \eta\, P_{hyd} \tag{8}$$

where η is the overall efficiency, which depends on wheel type, design, and operating condition. The torque and angular speed at the shaft are related by

$$P_{shaft} = \tau\, \omega \tag{9}$$

so that, for a given P_{shaft}, torque is **inversely** proportional to angular speed ($\tau = P_{shaft}/\omega$). A high-speed shaft carries low torque and a low-speed shaft carries high torque.

Gear ratio. Define the speed ratio $i = \omega_{stone}/\omega_{wheel}$. A speed-increasing gear train ($i > 1$) raises the millstone speed and, ignoring losses, lowers the torque in the same proportion:

$$\tau_{stone} = \eta_g\, \tau_{wheel} / i \tag{10}$$

where η_g is the gear-train efficiency. Vertical wheels turn slowly relative to the speed useful for grinding, so a speed-increasing gear train is typical for them. A horizontal-wheel mill can run directly ($i = 1$), which removes gear losses and simplifies the design. Reading “gear ratio” correctly therefore requires stating which shaft speed is the input and which is the output.

Illustrative example (hypothetical values, not site data). Let $Q = 0.05$ m³/s, $H = 3$ m, and $\eta = 0.5$. Then $P_{hyd} = 1000 \times 9.81 \times 0.05 \times 3 \approx 1.47$ kW and $P_{shaft} \approx 0.74$ kW. If the wheel turns at 10 rpm ($\omega \approx 1.05$ rad/s), $\tau_{wheel} \approx 700$ N·m. Through a gear train with $i = 12$ and $\eta_g = 0.9$, the millstone turns at 120 rpm ($\omega \approx 12.6$ rad/s) with $\tau_{stone} \approx 53$ N·m and delivered power ≈ 0.66 kW. The efficiency η is assumed here; published efficiencies for Roman vertical wheels [7], [8] should not be assumed to apply to the Ajloun mills.

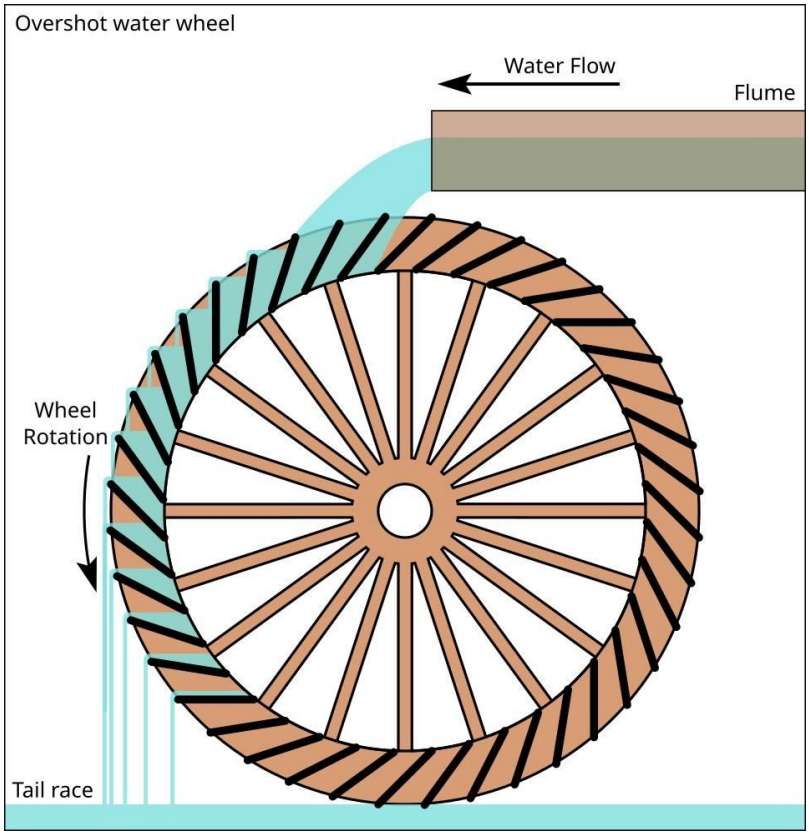


Figure 3. Generic overshot water wheel: water enters from the flume at the top and fills buckets that drive the wheel under gravity; spent water leaves at the tailrace. Shown to illustrate Eqs. (7)–(9); the wheel type at the Ajloun mills is not established in the sources reviewed. Source: [SOURCE AND LICENCE TO BE ADDED].

6.3 Critical assessment and evidence grade

The Ajloun mills have the weakest technical documentation of the three cases. The main scholarly source [5] covers the region broadly, and the site-specific material [6] is heritage documentation that is **not peer-reviewed**. Measured heads, discharges, wheel types, and gear details for individual mills are not available in the sources reviewed.

Evidence grade: C.

7. Discussion

7.1 Cross-case comparison

	Petra	**Jerash**	**Ajloun valley**
Principle	Free-surface critical flow; pipe hydraulics	Slider-crank kinematics	Energy conversion, power transmission
Period	1st c. BCE–3rd c. CE	6th c. CE	Mamluk–Ottoman (to early 20th c.)
Main sources	[1], [2]	[3], [4]	[5], [6]
Evidence grade	B	B (existence) / C (mechanics)	C
Main weakness	Single research programme; no experimental validation	No published dimensions or efficiency	Non-peer-reviewed site sources; wheel type unconfirmed

The three cases illustrate mechanical principles that are still taught today. They are separate examples from different periods, cultures, and technological contexts. The present evidence does not show transmission of knowledge between them, so no claim of a single continuous engineering tradition is made. Whether the builders worked by empirical iteration, inherited craft knowledge, or imported technology is a historical question that this review cannot settle.

7.2 Research gaps

1. **Experimental validation** of CFD reconstructions of Petra, for example through scaled physical models or measurements on surviving sections.
2. **Sensitivity analysis** of how strongly the Petra conclusions depend on assumed roughness, pipe diameter, leakage, and inflow.
3. **Independent analyses** by researchers other than the original author, and engagement with alternative interpretations of the gradients.
4. **Jerash:** publication of surviving dimensions, a kinematic and power reconstruction with explicit uncertainty, and comparison with Hierapolis.
5. **Ajloun:** a documented survey of wheel types, heads, discharges, and gearing, published in a peer-reviewed venue.
6. **Transferability of efficiency data** from Roman or modern wheels to the Levantine mills.

7.3 Limitations

This is a literature review and presents no original measurements. Its conclusions are only as strong as the sources it relies on, and the Petra evidence in particular rests on a small number of related publications. The two worked examples (Sections 5.2 and 6.2) use hypothetical parameters declared as such; they illustrate the equations and do not describe the sites. The review is purposive, not systematic, and unpublished or Arabic-language excavation reports may not have been captured. Efficiency data in references [7] and [8] concern Roman vertical-wheel mills and were not measured at the Jordanian sites.

8. Conclusion

Three cases from Jordan's history, the Petra water-supply system, the Jerash sawmill, and the Ajloun–Kufranjah watermills, illustrate mechanical principles that modern engineering expresses in terms of free-surface hydraulics, slider-crank kinematics, and energy conversion and power transmission. At Petra, published CFD work suggests that some conduit gradients are consistent with favourable hydraulic behaviour, though this rests on a narrow body of work and does not establish design intent. At Jerash, excavation shows a water-powered stone saw belonging to a wider Roman–Byzantine technology, but its technical parameters are unpublished. For the Ajloun mills, documentation is mostly non-peer-reviewed, and the wheel type at individual mills remains to be confirmed. The value of these sites for engineering history and education is real, but realizing it requires measured data, independent analysis, and peer-reviewed documentation.

Acknowledgments

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